

14 Temperature

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Temperature measures thermal state and determines direction of net thermal energy transfer. At absolute zero kelvin is zero, though particles still have quantum behaviour. A thermometric property must vary predictably with temperature; a calibrated scale turns property readings into temperatures. ΔT in kelvin equals ΔT in $^{\circ}\text{C}$.

Worked example: follow the reasoning

$27^{\circ}\text{C} \approx 300\text{ K}$. An increase of 15°C is an increase of 15 K, not 288 K.

Quick check · multiple choice

A Which conversion to kelvin is closest for 27°C ?

[1]

- A 27 K
- B 300 K
- C 246 K
- D 373 K

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A gas warms from 20°C to 80°C . Give initial and final absolute temperatures and temperature rise in kelvin.

[3]

Answers and explanation

Quick check: B. 300 K

Question 1 [3 marks]

Initial = 293 K [1]; final = 353 K [1]; rise = 60 K [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.